# Probability

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## 1. Probability of Drawing a Red Ball

A bag contains:

- Red balls = **3**
- Blue balls = **7**
- Total balls = **10**

### Formula:

P(E)=Favourable OutcomesTotal OutcomesP(E)=\frac{\text{Favourable Outcomes}}{\text{Total Outcomes}}

Here:

Favourable outcomes = 3 red balls

Total outcomes = 10 balls

P(Red)=310P(\text{Red})=\frac{3}{10}

### Answer:

P(Red)=0.3=30%\boxed{P(\text{Red})=0.3=30\%}

Therefore, the probability of drawing a red ball is **30%**.

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# 2. Tossing a Fair Coin 20 Times

A fair coin has two possible outcomes:

**Heads or Tails**

Suppose the coin is tossed 20 times and the following result is obtained:

**H, T, H, H, T, H, T, T, H, H, T, H, T, H, H, T, H, T, H, T**

Number of Heads = **11**

Total tosses = **20**

### Experimental Probability

P(Heads)=Number of HeadsTotal TossesP(\text{Heads})= \frac{\text{Number of Heads}}{\text{Total Tosses}} P(Heads)=1120P(\text{Heads})=\frac{11}{20} P(Heads)=0.55P(\text{Heads})=0.55

### Answer: 55%

**Note:** Since this is an experiment, the result can be different each time. The theoretical probability of Heads for a fair coin is **50%**.

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# 3. Probability Using a Standard Dice

A standard dice has **6 sides**:

**1, 2, 3, 4, 5, 6**

## A. Probability of Rolling an Even Number

Even numbers are:

**2, 4, 6**

Favourable outcomes = **3**

Total outcomes = **6**

P(Even)=36P(\text{Even})=\frac{3}{6} P(Even)=12=50%\boxed{P(\text{Even})=\frac{1}{2}=50\%}

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## B. Probability of Rolling a Number Greater Than 4

Numbers greater than 4 are:

**5, 6**

Favourable outcomes = **2**

Total outcomes = **6**

P(Number > 4)=26P(\text{Number > 4})=\frac{2}{6} P(Number > 4)=13≈33.33%\boxed{P(\text{Number > 4})=\frac{1}{3}\approx33.33\%}

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# 4. Probability Using a Deck of 52 Cards

A standard deck contains **52 cards**.

## A. Probability of Drawing an Ace

There are **4 Aces** in a deck:

- Ace of Hearts
- Ace of Diamonds
- Ace of Clubs
- Ace of Spades

Favourable outcomes = **4**

Total outcomes = **52**

P(Ace)=452P(\text{Ace})=\frac{4}{52}
P(Ace)=113≈7.69%\boxed{P(\text{Ace})=\frac{1}{13}\approx7.69\%}

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## B. Probability of Drawing a Red Card Given That It Is a Face Card

Face cards are:

- Jack
- Queen
- King
- Ace

There are 4 such cards in each suit.

Therefore:

**Total face cards = 4 × 4 = 16**

Red suits are:

- Hearts
- Diamonds

Each has 4 face cards.

Therefore:

**Red face cards = 8**

### Conditional Probability:

P(Red | Face Card)=Red Face CardsTotal Face CardsP(\text{Red | Face Card}) = \frac{\text{Red Face Cards}}{\text{Total Face Cards}} =816=\frac{8}{16} P(Red | Face Card)=12=50%\boxed{P(\text{Red | Face Card})=\frac{1}{2}=50\%}

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# 5. Probability That a Pizza Order Was Placed at Night

Given:

- Probability of Night order = **60% = 0.60**
- Probability of Day order = **40% = 0.40**
- Probability of Pizza given Night = **30% = 0.30**
- Probability of Pizza given Day = **10% = 0.10**

We need to find:

### P(Night | Pizza)

Using conditional probability:

P(Night∣Pizza)=P(Night)×P(Pizza∣Night)P(Night)×P(Pizza∣Night)+P(Day)×P(Pizza∣Day)P(Night|Pizza) = \frac{P(Night)\times P(Pizza|Night)} {P(Night)\times P(Pizza|Night)+P(Day)\times P(Pizza|Day)}

Substituting the values:

=0.60×0.30(0.60×0.30)+(0.40×0.10)=\frac{0.60\times0.30} {(0.60\times0.30)+(0.40\times0.10)} =0.180.18+0.04=\frac{0.18}{0.18+0.04} =0.180.22=\frac{0.18}{0.22} =0.8182=0.8182

### Answer: 81.82%

Therefore, the probability that a randomly selected **pizza order was placed at night is approximately 81.82%**.

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# Final Summary

| Question | Answer |
|---|---|
| Red ball probability | **30%** |
| Experimental probability of Heads | **55%** |
| Probability of even number on dice | **50%** |
| Probability of number greater than 4 | **33.33%** |
| Probability of drawing an Ace | **7.69%** |
| Probability of red card given face card | **50%** |
| Probability pizza order was placed at night | **81.82%** |